ModelIt!

Student Activity Pack

Why Do Some Animals Look Different?

Inherited Traits vs. Learned Behaviors

Grade 3 | NGSS 3-LS3-1 | Patterns

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Do you look more like your mom or your dad? What features did you get from each?

2 Why do you think brothers and sisters from the same family do not look exactly the same?

3 Draw two parent animals and what you think their baby might look like.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Parent Traits | Genetic Variation | Offspring Appearance

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Similar Parents

Parent Traits set to very similar

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Different Parents

Parent Traits set to very different

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Mixed Parents

Parent Traits set to somewhat different

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Environmental Factors — Things in the environment that change how an organism grows — like sunlight, food quality, and temperature

Number of Offspring — How many babies are born — more offspring means more chances to see trait variation in the group

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do some puppies from the same litter look different from each other? Use evidence from your simulation to support your answers.

1 What is the difference between a trait you inherited and something you learned or got from your environment?

2 How did your model help you understand why no two animals (or people) are exactly alike?

8 STEM Challenge: Design a Trait Prediction Game

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YOUR ENGINEERING MISSION

Create a game that shows how mixing parent traits produces unique offspring with different combinations of features.

THE SCENARIO

A children's science museum wants a fun, interactive game that teaches kids about inherited traits. Your team must design a game where players combine parent trait cards to predict what their offspring will look like. Use what you learned about traits and variation!

Engineering Design Thinking

1. HOW MANY TRAITS WILL EACH PARENT CARD HAVE?

2. HOW WILL PLAYERS COMBINE PARENT CARDS TO CREATE OFFSPRING — FLIPPING COINS, ROLLING DICE, DRAWING CARDS?

3. HOW WILL YOU DRAW OR SHOW WHAT THE OFFSPRING LOOKS LIKE BASED ON THE COMBINED TRAITS?

4. CAN YOU MAKE IT SO NO TWO OFFSPRING ARE EXACTLY THE SAME?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Geneticists study how traits are passed from parents to offspring and why living things look the way they do. They earn $55,000-$100,000/year.